Department of Nano Science Technology
Faculty of Technology
Wayamba University of Sri Lanka

Centralized Workshops & Lab Scheduling Platform

Database SQL Query Sheet Database SQL Query Sheet

Submitted to: Dr. U.S. Liyanarachchi
Module: NANO2142

Submitted by: Group 1

Date of Submission

Sep 03 2026

Database SQL Query Sheet: Student Workshop Booking System

1. Data Definition Language (DDL) - Database Schema Creation The following queries define the relational database structure, including primary keys, foreign keys, and relationships between the four main entities: STUDENTS, WORKSHOPS, BOOKINGS, and ADMIN_LOGS.

```
-- 1.1 Create STUDENTS Table
CREATE TABLE STUDENTS (
   student_id SERIAL PRIMARY KEY,
   index_no VARCHAR(50) UNIQUE NOT NULL,
   name VARCHAR(100) NOT NULL,
   department VARCHAR(100),
   email VARCHAR(150)
);

-- 1.2 Create WORKSHOPS Table
CREATE TABLE WORKSHOPS (
   workshop_id VARCHAR(50) PRIMARY KEY,
   category_name VARCHAR(100) NOT NULL,
   description TEXT,
   icon_path VARCHAR(255)
);

-- 1.3 Create BOOKINGS Table
CREATE TABLE BOOKINGS (
   booking_id VARCHAR(50) PRIMARY KEY,
   student_id INT NOT NULL,
   workshop_id VARCHAR(50) NOT NULL,
   booking_date DATE NOT NULL,
   time_slots VARCHAR(255) NOT NULL,
   pdf_path VARCHAR(255),
   description TEXT,
   passcode VARCHAR(10) NOT NULL,
   status VARCHAR(50) DEFAULT 'Pending',
   created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
   FOREIGN KEY (student_id) REFERENCES STUDENTS(student_id) ON DELETE
CASCADE,
   FOREIGN KEY (workshop_id) REFERENCES WORKSHOPS(workshop_id) ON
DELETE CASCADE
);

-- 1.4 Create ADMIN_LOGS Table
CREATE TABLE ADMIN_LOGS (
   log_id SERIAL PRIMARY KEY,
   booking_id VARCHAR(50) NOT NULL,
   action_type VARCHAR(100) NOT NULL,
   action_timestamp TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
   FOREIGN KEY (booking_id) REFERENCES BOOKINGS(booking_id) ON DELETE
CASCADE );
```

2. Data Manipulation Language (DML) - Data Insertion The following queries demonstrate how initial data is inserted into the respective tables.

```
-- 2.1 Insert Student Data
INSERT INTO STUDENTS (index_no, name, department, email)
VALUES ('555555', 'Sachintha Rashmika', 'Technology', 'ppas.rashmika@example.com');

-- 2.2 Insert Workshop Category Data
INSERT INTO WORKSHOPS (workshop_id, category_name, description, icon_path)
VALUES ('WKS-AUTO-01', 'Automotive', 'Vehicle systems and maintenance.',
'/assets/automotive.png');

-- 2.3 Insert a New Booking Record
INSERT INTO BOOKINGS (booking_id, student_id, workshop_id, booking_date,
time_slots, pdf_path, description, passcode, status)
VALUES ('BKG-260903120', 1, 'WKS-AUTO-01', '2026-09-03', '8.30 am - 10.30 am',
'/uploads/pdfs/BKG-260903120.pdf', 'Need for final year project.', '4582', 'Pending');

-- 2.4 Insert an Admin Action Log
INSERT INTO ADMIN_LOGS (booking_id, action_type)
VALUES ('BKG-260903120', 'Status Updated to Approve');
```

3. Data Query Language (DQL) - Data Retrieval & Joins These queries are used to extract meaningful information from multiple tables using relationships.

```
-- 3.1 Retrieve all bookings with comprehensive Student and Workshop details
SELECT
   b.booking_id,
   s.name AS student_name,
   s.index_no,
   w.category_name AS workshop,
   b.booking_date,
   b.time_slots,
   b.status
FROM BOOKINGS b
INNER JOIN STUDENTS s ON b.student_id = s.student_id
INNER JOIN WORKSHOPS w ON b.workshop_id = w.workshop_id
ORDER BY b.created_at DESC;

-- 3.2 Filter bookings for a specific calendar date
SELECT * FROM BOOKINGS
WHERE booking_date = '2026-09-03';

-- 3.3 Retrieve the administrative action history for a specific booking
SELECT action_type, action_timestamp
FROM ADMIN_LOGS
WHERE booking_id = 'BKG-260903120'
ORDER BY action_timestamp ASC;
```

4. Data Modification & Deletion (Update & Delete Queries) Queries used by the system administrators to manage existing booking records.

```
-- 4.1 Update Booking Status (e.g., Approve or Reject)
UPDATE BOOKINGS
SET status = 'Approve'
WHERE booking_id = 'BKG-260903120';

-- 4.2 Update a Student's Email Address
UPDATE STUDENTS
SET email = 'updated.email@example.com'
WHERE index_no = '555555';

-- 4.3 Delete a Booking Record (Cascades to remove related Admin Logs)
DELETE FROM BOOKINGS
WHERE booking_id = 'BKG-260903120';
```